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Road users' stereotypes, emotions and behavioral intentions: Comparisons between motorists, cyclists and pedestrians

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ABSTRACT

This study examines social judgments among road users by focusing on how motorists, cyclists, and pedestrians perceive both their own group and others based on their mode of transportation. Drawing on the Stereotype Content Model (SCM) and the Behavior from Intergroup Affect and Stereotype (BIAS) map, the stereotypes, emotions, and behavioral intentions associated with each group were analyzed (motorists, n = 124; cyclists, n = 114; pedestrians, n = 137). Results show that mode of transport serves as a key social marker in shaping group identities, with vulnerable road users (cyclists and pedestrians) demonstrating strong in-group favoritism, in contrast to motorists. Through the BIAS map, a detailed mapping of the emotions and behaviors directed toward each group is offered, and a perceptual system is proposed that captures how these three social groups relate to one another in traffic contexts. This framework provides a valuable tool for understanding the dynamics of intergroup perception within a complex system of road users. Moreover, these implicit biases—often automatic and unconscious—are emphasized as potential causes of conflictual interactions and contributors to serious, even fatal, accidents. Findings finally suggest that addressing such biases could improve interactions among road users and enhance overall traffic safety.

1. Introduction

The growing number of pedestrians and cyclists, driven by pro-environmental and health policies, is leading to new forms of interaction among these vulnerable road users and individual vehicle drivers (Gilderbloom et al., 2016). These increasing interactions between members of these three road-user groups result in heightened conflicts and a greater risk of accidents, raising significant concerns for public health and road safety (Kaplan & Prato, 2015). Addressing these issues could both prevent accidents and promote peaceful coexistence between different groups of road users in shared spaces. For instance, in the French context of the present study, cyclists experienced an 8 % increase in the number of severe road injuries between 2019 and 2023 (French National Interministerial Road Safety Observatory, 2023). This trend is particularly pronounced in urban areas where interactions and conflicts between users are more frequent. In line with this, the present study focuses on motorists, cyclists, and pedestrians, who together represent the overwhelming majority (95.9 %) of commuting trips in France according to the 2022 population census: 71 % by car, 21.4 % by walking (including trips via public transport), and 3.5 % by bicycle (French National Institute of Statistics and Economic Studies, 2022).

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A large body of research in intergroup relations within social psychology has demonstrated that representations of groups (i.e., stereotypes) are significant predictors of how people will perceive, feel, and act toward others based on their group membership (Bodenhausen et al., 2012; Fiske, 2018; Yzerbyt & Demoulin, 2010). When encountering a member of a particular group, information about that group is implicitly activated (Ferguson & Bargh, 2004), including attitudes, beliefs and expectations. Depending on the content of the stereotype, this activation may lead to different forms of prejudiced or biased outcomes (e.g., neglect, contempt, discrimination, aggression; Cuddy et al., 2007, 2009). Applying this knowledge to the traffic context, where distinct groups of users interact, is crucial for understanding how automatically activated representations may lead to biased behaviors toward individuals based on their group membership, potentially resulting in conflicts or accidents. Nonetheless, to this day, relatively little is known about how perceptions of various road users in relation to each other may influence interactions on the road (Goddard, 2022).

The objective of the present study is to offer a more exhaustive understanding of how the three main groups of road users (motorists, cyclists and pedestrians), who may encounter various conflictual situations on the road, perceive and represent each other, and how these perceptions may influence their behavioral intentions towards one another. To that end, social-cognitive models of human perception were used: the Stereotype Content Model (SCM, Fiske et al., 2002) and its extension, the Behavior from Intergroup Affect and Stereotype map (BIAS map, Cuddy et al., 2007). These two models, to the best of our knowledge, have not been previously used in traffic psychology research while they have provided valuable insights in applied research for predicting behavior or behavioral intentions across various fields, including marketing (Guo et al., 2022; Kervyn et al., 2012), wildlife conservation (Sevillano & Fiske, 2019a), and healthcare (Boysen, 2017; Reihl et al., 2015). For instance, in the healthcare domain, existing stereotypes about mental disabilities shape medical practices, such as decisions regarding institutionalization (Reihl et al., 2015). Likewise, the dimensions of the SCM provide insights into the quality of patient-caregiver relationships, influenced by the caregivers' social backgrounds (Jain, 2022).

1.1. Social perception of road users

The road environment acts not only as a space for transit but also as a context in which transportation modes are salient, contributing to the formation and expression of social identities (Aldred, 2013). Past research provided evidence that the type of vehicle (e.g., private car, bike) serves as a relevant social cue for identity formation and group perception (Goddard et al., 2020; Hoekstra et al., 2018; Prati et al., 2017). Motorists, for instance, tend to identify more strongly with other car drivers than with cyclists, and vice versa (Hoekstra et al., 2018). This identification extends to expectations of behavior in conflictual situations, with motorists anticipating more leniency from other drivers than from cyclists, and the reverse is also true. Social group dynamics thus seem to play a role on the road, with various users interacting and perceiving one another. However, the nature of these perceptions remains somewhat unclear and the current literature lacks evidence and a unified framework to understand those perceptions and their implications for road users. For example, there seems to be a stereotype among motorists of cyclists as recurrent lawbreakers who have a poor understanding of road rules and often engage in risky behavior (Aldred, 2013; Basford et al., 2002). However, less is known about how these stereotypes affect motorists' perceptions and behavioral intentions towards cyclists as a group. Similarly, limited information is available on how cyclists perceive motorists. Conversely, cyclists may perceive pedestrians as inattentive and discourteous (Davies et al., 2003; Paschalidis et al., 2016), while viewing fellow in-group cyclists as assertive and well-informed (Gatersleben & Haddad, 2010). Additionally, cyclists tend to perceive pedestrians as more vulnerable, thus their interaction may differ from that with vehicles (Paschalidis et al., 2016).

Finally, motor vehicles are given predominance in overall traffic, with rules and infrastructure shaped to their advantage, often at the expense of other user groups such as cyclists and pedestrians (Goddard, 2022; Prati et al., 2017). This competitive asymmetry in the road environment may contribute to biased perceptions of user groups, influenced by the perspectives of both dominant groups and

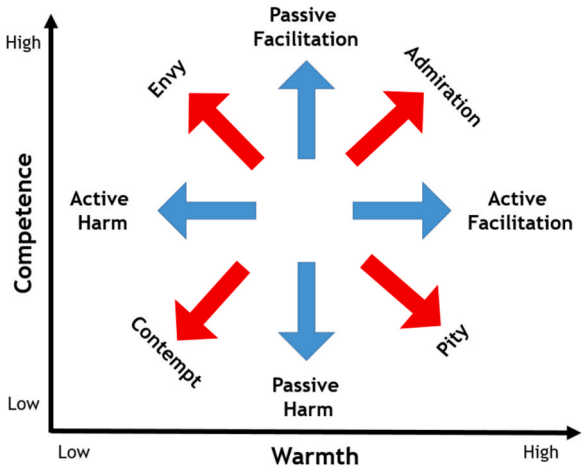

Fig. 1. Behavioral intentions from intergroup affect and stereotype (adapted from Cuddy et al., 2007).

minorities such as pedestrians and cyclists. These biases further shape how different user groups perceive and interact with one another (Fiske & Durante, 2019).

1.2. The stereotype content model and the behaviors from intergroup affect and stereotype map

The Stereotype Content Model (SCM; Fiske et al., 2002) describes the content of stereotypical beliefs about diverse categories of people (e.g., societal subgroups, ethnicities, sexual orientations). The SCM proposes that social perception is organized along two fundamental dimensions: warmth and competence (Cuddy et al., 2009; Fiske et al., 2007). Warmth reflects perceived intentions toward others (e.g., friendliness, trustworthiness vs. hostility, malevolence), while competence pertains to perceived abilities, skills, and efficacy in achieving goals or tasks. These dimensions form a two-dimensional space where individuals and groups are evaluated based on their perceived warmth and competence. Evidence suggests that these dimensions are present across cultures, with some social groups eliciting universally shared perceptions (Cuddy et al., 2009; Fiske & Durante, 2019). For instance, immigrants and the homeless tend to be perceived negatively, as low in warmth and competence (negative stereotype), while middle-class citizens are viewed more positively, as high in both warmth and competence (positive stereotype). Some groups evoke ambivalent stereotypes: technical experts are seen as highly competent but low in warmth (envious stereotype), whereas the elderly are perceived as high in warmth but low in competence (paternalistic stereotype).

The SCM later evolved into the BIAS map (Behaviors from Intergroup Affect and Stereotype), which incorporates emotions and behavioral intentions toward groups based on stereotypical content (Cuddy et al., 2007, 2008). As shown in Fig. 1, the BIAS map posits that perceptions of a group's warmth and competence trigger distinct emotions. Positively perceived groups (high warmth and competence) elicit admiration, while negatively perceived groups (low warmth and competence) elicit contempt. Ambivalently perceived groups evoke mixed emotions: those high in competence but low in warmth are envied, whereas those high in warmth but low in competence elicit pity. Stereotypes and emotions, in turn, predict behavioral intentions, characterized by their valence (facilitation vs. harm) and intensity (active vs. passive). Perceptions of warmth primarily drive active behaviors (i.e., direct, explicit, confrontational), as warmth cues create a pressing need to act (Cuddy et al., 2007). In contrast, competence cues are more predictive of passive behaviors (i.e., indirect, covert, avoidant; Cuddy et al., 2007). Consequently, admired groups (high warmth and competence) elicit both passive and active facilitation (e.g., cooperation and protection) while disdained groups (low warmth and competence) provoke both passive and active harm (e.g., neglect and aggression). Envied groups (high competence, low warmth) elicit ambivalent responses: passive facilitation (e.g., cooperation) but active harm (e.g., aggression). Inversely, pitied groups (high warmth, low competence) evoke active facilitation (e.g., protection) alongside passive harm (e.g., neglect).

1.3. The present study

As interactions between motorists, cyclists, and pedestrians increase, so do the opportunities for conflicts and accidents. The SCM and BIAS map models, when applied to road traffic, may offer valuable insights into the factors influencing conflict and accident-related behaviors among road users. Unlike traditional SCM-BIAS map studies, which explore broad cultural stereotypes about different social groups (i.e., how society in general perceives a target group), the present research investigates specific stereotypes that distinct groups of road users hold towards one another (e.g., how motorists perceive cyclists, pedestrians, and other motorists, and vice versa).

We expect that individuals' perceptions of other road-user groups will vary based on their own group membership. As discussed in the introduction, evidence suggests that the mode of transport individuals use contributes to distinct social categories on the road. Furthermore, research on social perception shows that the framing of questions significantly influences responses. For example, asking participants about societal perceptions tends to elicit different answers than asking for their personal opinions (Findor et al., 2020; Kotzur et al., 2020). In the present study, members of each group (motorists, cyclists, pedestrians) were asked how they perceive both in- and out-group members. This method helps to reduce social desirability bias (by not asking for personal opinions) while providing useful insights into group-specific perceptions (by not asking for societal points of view), making it a suitable methodological approach for our research questions.

Hypotheses one and two (H1 and H2) in this study are derived directly from the SCM and the BIAS Map, aiming to assess whether these models are applicable in the context of road users. Investigating these hypotheses allows us to explore the assumption that different modes of transportation shape perceptions of road user groups, eliciting distinct representations (stereotypes) and associated emotional responses.

H1 (stereotypes → behaviors): The perceived warmth of road-user groups will be positively associated with active facilitation and negatively associated with active harm. The perceived competence of road-user groups will be moreover positively associated with passive facilitation and negatively associated with passive harm.

H2 (emotions → behaviors): Admiration will be positively associated with both active and passive facilitation. Contempt will be positively associated with both active and passive harm. Pity will be positively associated with active facilitation and passive harm, while envy will be positively associated with passive facilitation and active harm.

Regarding in- and out-group dynamics, we expect in hypothesis 3 that road users will have more favorable representations, experience more positive emotions (and less negative ones), and exhibit more positive behavioral intentions (i.e., more facilitation) toward in-group members compared to their judgments, emotions, and intentions (i.e., more harm) toward out-group members. This prediction is in accordance with previous results from Hoekstra et al. (2018) and social identity theory (Tajfel & Turner, 2001) indicating that people usually show in-group favoritism.

H3 (in-group favoritism): Individuals' representations of their own road-user group will be more positive on the dimensions of warmth and competence compared to their perceptions of other road-user groups. Additionally, emotions toward in-group members will be more positive (i.e., more admiration), alongside greater behavioral intentions toward facilitation, whereas emotions toward out-group members will be more negative (i.e., more contempt) with a greater inclination towards harmful behaviors.

Finally, our research will allow us to map how each group perceives one another in the road environment. Since this aspect is primarily exploratory, we do not have any hypotheses regarding how each group judges the others based on the SCM-BIAS map.

2. Method

The materials and data supporting the study findings are available online in a file on the Open Science Framework (OSF) at the following link: https://osf.io/msf3d/.

2.1. Sample size rationale

Sample size calculation was based on the following design: 3, reported user group (between-subject; motorist vs. cyclist vs. pedestrian) × 3, user group evaluation (within-subject; motorists, cyclists, pedestrians) on GPower (version 3.1.9.7). This specification reflected the procedure of the study, in which each participant evaluated all three groups but belonged to only one self-reported user group, a common approach in intergroup perception research. Assuming a small effect size ($f = .10$), $\alpha = .05$, and power = .95,[1] GPower estimated the minimum required sample at 312 participants (104 per group). G*Power computes this value by determining the non-central F-distribution under the specified parameters, ensuring sufficient sensitivity to detect even subtle effects.

2.2. Participants

A total of 390 participants took part in the study and completed their participation. Fifteen participants were excluded as they failed at least one of the attention-check questions. The final sample consisted of 375 participants living in France (42.4 % female, 56.8 % male, 0.8 % other/not indicated), ranging from 18 to 88 years old ($M = 34$, $SD = 11.1$). Participants were recruited either from the crowdsourcing platform Prolific Academic ($n = 286$, 76.3 %) or from mailing lists ($n = 86$, 23.7 %). Among the participants, $n = 124$ indicated that they were motorists, $n = 114$ cyclists, and $n = 137$ pedestrians. For their daily commutes, motorists reported using their cars an average of 5.14 days per week ($SD = 1.43$), cyclists rode their bicycles an average of 5.65 days per week ($SD = 1.24$), and pedestrians reported walking an average of 6.09 days per week ($SD = 1.33$). Only a minority of participants indicated that they were a member of an organization that advocates for cyclists or pedestrian road users ($n = 8$ for pedestrians; $n = 20$ for cyclists). This variable was recorded for potential control purposes given that activists may hold more polarized attitudes toward in- and out-group members (Kutlaca et al., 2020). Full demographic details are provided in Table 1.

2.3. Procedure

Participants completed an online questionnaire which was described as a survey measuring road users' perception of different types of other road users. Traditionally, in most studies using the SCM-BIAS map, each participant rates a certain number of randomly presented social groups (e.g., Bye & Herrebrøden, 2018). Therefore, in the present study, each participant rated the three groups of motorists, cyclists, and pedestrians presented in a random order. Our first variable, "reported user group," is a between-subject factor determined by the participants' self-reported main mode of transportation. Participants were asked "In general, in your daily life, what is your primary means of transportation?" (motorist, cyclist, or pedestrian). This self-categorization approach, consistent with previous research in road-user perception, provides a meaningful basis for group identification (Hoekstra et al., 2018). Our second variable, "group judgment," is a within-subject factor, each participant evaluating all three groups of road users (motorists, cyclists and pedestrians).

Overall, the questionnaire comprised 72 items, corresponding to 24 items for each of the three judged groups (motorists, cyclists, pedestrians). These items consisted of the 12 two-item subscales of the SCM-BIAS map, assessing stereotype content, emotions, and behavioral intentions (further detailed in section 2.4 Material). An additional 12 items collected participants' primary mode of transportation, demographic characteristics, and included two attention-check items to ensure data quality.

Ethics approval was obtained from the university's ethics review board prior to data collection.

2.4. Material

Participants first indicated their primary mode of transportation by responding to the question: "In general, in your daily life, what is your primary means of transportation?", similarly to Hoekstra et al. (2018). Subsequently, participants completed the SCM-BIAS map subscales to assess their judgments of three groups of road users (motorists, cyclists, pedestrians), presented in a random order. Questions were framed in such a way that participants responded from the perspective of their selected mode of transportation

[1] As there is no pilot study available for this experimental protocol, default recommended values were utilized for the correlation among repeated measures ($r = 0.5$) and for non-sphericity correction ($\varepsilon = 1$).

Table 1
Demographic details of the sample.

	Motorists	Cyclists	Pedestrians
Sample size	$n = 124$	$n = 114$	$n = 137$
Mean age (*SD*)	35.3 (10.9)	35.7 (11.0)	31.5 (11.0)
Gender	51F, 72 M, 1 Other	36F, 77 M, 1 Other	72F, 64 M, 1 Other
Highest diploma[a]	Master's degree (43 %)Bachelor's degree (29 %)	Master's degree (57 %)PhD (21 %)	Master's degree (48 %)Bachelor's degree (25 %)
Professions[a]	Executive positions (35 %)Employee (29 %)	Executive positions (59 %)Employee (17 %)	Employee (26 %)Student (26 %)
Advocacy groups[b]	$n = 3$ (cyclists)	$n = 17$ (cyclists)	$n = 7$ (pedestrians) $n = 1$ (cyclists)
Urban residency	50 %	90 %	95 %
Car usage[c]	5.14 ($SD = 1.43$)	0.54 ($SD = 0.77$)	0.77 ($SD = 1.03$)
Bicycle usage[c]	0.64 ($SD = 1.43$)	5.65 ($SD = 1.24$)	0.82 ($SD = 1.44$)
Walking[c]	2.53 ($SD = 1.98$)	2.61 ($SD = 2.07$)	6.09 ($SD = 1.33$)

Note. [a]Only the two most represented categories are displayed. [b]Terms in parentheses indicate which group participants are advocating for. [c]Measured in days/week.

(e.g., how do cyclists perceive motorists). While completing the SCM-BIAS map subscales for each road user group judged, participants viewed an image depicting themselves from the viewpoint of their selected mode of transportation, sharing the road with members of the judged group. This visualization aimed to situate them concretely in a traffic context involving road space sharing with the judged group. These images are accessible on the OSF platform file.

For each group judged, the SCM-BIAS Map comprised two-item subscales measuring perceived warmth, competence, emotions (admiration, pity, contempt, envy), and behavioral tendencies (active facilitation, passive facilitation, active harm, passive harm).[2] The final questionnaire contained 24 items per judged group, resulting in a total of 72 items.

The item translation into French and adaptation to the traffic environment underwent several validation steps, all detailed in Appendix A. The final items used in each two-item subscale were as follows[3]: *warm* and *friendly* (warmth), *competent* and *capable* (competence), *contempt* and *disgust* (contempt), *pity* and *sympathy* (pity), *admire* and *respect* (admiration), *help* and *protect* (active facilitation), *fight* and *attack* (active harm), *cooperate* and *associate with* (passive facilitation), *exclude* and *hinder* (passive facilitation). All items were rated on a 7-point Likert scale (1 = 'not at all', 7 = 'absolutely').

To assess the internal consistency of the SCM-BIAS subscales, Cronbach's α was computed separately for each judged group and then averaged across groups. As each subscale consisted of two items, reliability estimates were based on these item pairs. Reliabilities for the SCM-BIAS map two-item scales were as follows: competence $\alpha = .93$; warmth $\alpha = .91$; admiration $\alpha = .62$; envy $\alpha = .81$; contempt $\alpha = .80$; pity $\alpha = .53$; active facilitation $\alpha = .82$; passive facilitation $\alpha = .86$; active harm $\alpha = .88$ and passive harm $\alpha = .59$. Due to questionable reliability for the admiration, pity, and passive harm subscales, we used only one of the most representative items from each of these two-item subscales in analyses, namely the "admiration" item for admiration, the "pity" item for pity, and the "ignore" item for passive harm.

The questionnaire included two attention-check measures in the format: "Please select '7 – absolutely' to show that you are paying attention." Additionally, a "coherency" check measure was included, asking participants to report their main mode of transportation again at the end of the study to ensure consistency. The survey also collected demographic information and data on the number of days per week participants used a car, bicycle, or walking as their primary means of transportation for basic activities as well as membership of an organization advocating for groups of road users. The online survey took approximately 15 min to complete.

3. Results

Statistical analyses were conducted on R statistical software (R version 4.3.3).

Demographic variables (age, gender, frequency of use of the transportation mode in question, affiliation with a road user advocacy group, and urban versus non-urban residency) did not show significant effects in any of the hypothesis-testing models.

3.1. Applicability of the SCM-BIAS map models to the traffic environment

The correlations among SCM-BIAS map measures were computed in two ways and can be found in Table 2 and Table 3. First, at the

[2] Two complementary subscales were added to measure perceived agency and morality —dimensions that supplement warmth and competence, respectively (Abele et al., 2016)— though these subscales were not part of the original SCM-BIAS Map (Cuddy et al., 2007).

[3] For the complementary added subscales, agency was measured using the items "self-confident" and "feel superior"; morality with "reliable" and "trustworthy." As these subscales were not part of the original BIAS map model, related analyses are reported in the supplementary material only.

Table 2
Correlations of behavioral tendencies with stereotypes and emotions for each of the three judged road-user groups.

Predictor	Behavioral tendency Active facilitation	Active harm	Passive facilitation	Passive harm
Judged group: motorists				
Stereotypes				
Competence	.25***	–.23***	**.33***	**–.14****
Warmth	**.44***	**–.12***	.37***	–.03
Emotions				
Admiration	**.39***	–.03	**.20***	.05
Contempt	–.22***	**.35***	–.23***	**.17***
Envy	.12*	**.14****	**.08**	.09
Pity	**.18***	.11*	–.00	**.06**
Judged group: cyclists				
Stereotypes				
Competence	.44***	–.52***	**.53***	**–.15****
Warmth	**.57***	**–.51***	.60***	–.24***
Emotions				
Admiration	**.45***	–.23***	**.43***	–.16**
Contempt	–.38***	**.67***	–.51***	**.34***
Envy	.12*	**–.04**	**.17***	.03
Pity	**.11***	.18***	–.03	**.10***
Judged group: pedestrians				
Stereotypes				
Competence	.41***	–.37***	**.38***	**–.11***
Warmth	**.40***	**–.30***	.35***	–.12*
Emotions				
Admiration	**.22***	–.07	**.19***	–.04
Contempt	–.20***	**.53***	–.26***	**.30***
Envy	–.03	**.19***	**–.00**	.11*
Pity	**.03**	.26***	–.03	**.13***

Note. ***$p < .001$, **$p < .01$, *$p < .05$; Bold correlations were predicted to be significant (31 out of 36 were). However, 25 out of 36 other ones were also significant, but they were theoretically consistent and this trend is common in studies using SCM-BIAS maps (e.g., Bye & Herrebrøden, 2018; Cuddy et al., 2007; Sevillano & Fiske, 2019b).

Table 3
Correlations of behavioral tendencies with stereotypes and emotions at the participant level.

Predictor	Behavioral tendency Active facilitation	Active harm	Passive facilitation	Passive harm
Participant level				
Stereotypes				
Competence	.65***	–.67***	**.49***	**.08**
Warmth	**.81***	**–.74***	.57***	.09
Emotions				
Admiration	**.75***	–.61***	**.62***	.02
Contempt	–.76***	**.62***	–.63***	**.05**
Envy	–.19**	**.22***	**.11**	–.16*
Pity	**.63***	–.41***	.33***	**.17****

Note. ***$p < .001$, **$p < .01$, *$p < .05$; Bold correlations were predicted to be significant (9 out of 12 were).

"transport-group level," we averaged the ratings across participants for each of the three judged groups, resulting in three distinct correlation matrices.[4] Second, at the "participant level," we followed the methodology from Cuddy et al. (2007): we calculated correlations separately for each participant, transformed them using Fisher's r-to-z transformation,[5] averaged them, and then converted them back to r.

Hypothesis 1. *(stereotypes → behaviors).* As predicted and as displayed in Table 2 and Table 3, warmth was positively correlated with active facilitation and negatively correlated with active harm across the three judged groups, as well as at the participant level. Competence was significantly and positively correlated with passive facilitation and negatively correlated with passive harm across the three judged groups of road users and at the participant level, with one exception: competence did not significantly correlate with

[4] This approach was preferred over the traditional "group-level" correlation computation used in the SCM-BIAS map literature, as our study involved only the rating of three groups. Applying the classic methodology would have resulted in insufficient statistical power for correlations at this level, as correlations would be calculated based on $n = 3$.

[5] The Fisher transformation is defined as $z = 0.5 \times ln(\frac{1+r}{1-r})$. The inverse transformation is $= \frac{e^{2z}-1}{e^{2z}+1}$.

passive harm at the participant level. These correlations provide sufficient confidence to conclude that there is a significant link between stereotypes and behavioral tendencies among road users, thus validating our first hypothesis.

Hypothesis 2. (*emotions → behaviors*). Results indicate that admiration toward a group of road users was consistently and positively related to both active and passive facilitation across all three judged groups, as well as at the participant level. Similarly, contempt toward these groups was significantly and positively associated with both active and passive harm, across all user groups and at the participant level, except for the relationship between passive harm and contempt at the participant level. Regarding ambivalent emotions, the results were more nuanced. As predicted, envy toward the evaluated road-user groups was significantly and positively correlated with active harm in two out of three groups, as well as at the participant level. Thus, three of the four predicted correlations were confirmed. However, envy was only significantly and positively correlated with passive facilitation in the case of the judgement of cyclists, confirming only one of the four predicted correlations. Pity, as expected, was significantly associated with active facilitation in the evaluation of two groups (motorists and pedestrians) and at the participant level. Moreover, as expected, pity was also positively and significantly correlated with passive harm, particularly in two group evaluations (cyclists and pedestrians) and at the participant level.

Taken together, these results offer partial support for Hypothesis 2. Uncertainties remain only in the associations between envy and behaviors linked to active harm and passive facilitation. Nevertheless, the overall pattern of findings is both consistent and compelling.

3.2. In- and out-group representations and behavioral intentions

Linear models were used to analyze the interaction between participants' self-reported road-user group and their perceptions of others. Table 4 present interaction parameters and Fig. 2 the graphical results. Statistically significant interaction coefficients were found across all stereotype content subscales and all behavioral intention subscales. Post-hoc analyses were further conducted to detail the effect of group membership on road-users' group perceptions, using Holm-Bonferroni correction.

Stereotype Content. Regarding competence and warmth, cyclist participants rated their ingroup as significantly more competent and warmer than outgroup members (motorists and pedestrians). Similarly, pedestrian participants also viewed their ingroup as more competent and warmer than both outgroups. However, contrary to our hypothesis, motorist participants did not perceive their ingroup as superior in these dimensions. Instead, they rated pedestrians as the most competent and warmest group, with no significant differences between themselves and cyclists.

Emotions. Concerning the positive emotion of admiration, cyclist participants felt significantly more admiration toward their ingroup members compared to the two outgroups of motorists and pedestrians. However, no significant ingroup-biased positive emotions were found among motorists and pedestrian participants. Regarding the negative emotion of contempt, cyclist participants indicated feeling more contempt towards motorists but did not express greater contempt for pedestrians compared to their ingroup. Pedestrian participants, in turn, reported feeling more contempt toward both outgroups (cyclists and motorists). Finally, contrary to our hypothesis, *motorist* participants indicated feeling more contempt toward their ingroup compared to cyclists and pedestrians.

Behavioral Intentions. *Cyclist* participants were more inclined to exhibit active and passive facilitation behaviors toward ingroup members (i.e., cyclists) than toward outgroup members (i.e., pedestrians and motorists). They also indicated a greater tendency for active harm toward both out-groups and more passive harm toward pedestrians specifically. Concerning *pedestrian* participants, a similar pattern emerged: they reported greater active facilitation and lower active harm behaviors toward their own group than toward outgroups (i.e., cyclists and motorists). However, unlike cyclists, pedestrians did not exhibit significant differences in passive facilitation across groups. Contrary to our hypothesis, they reported more passive harm toward their own group than toward motorists. In

Table 4
Results of the models testing the interactions between self-reported road-user group and target group judgment, and key post-hoc comparisons.

DV	Interaction	Ratings by cyclists[1]	Ratings by motorists[1]	Ratings by pedestrians[1]
Competence	$F(4, 1124) = 17.5^{***}$	Cyclists > Pedestrians*	Motorists < Pedestrians**	Pedestrians > Motorists ***
		Cyclists > Motorists***	Motorists = Cyclists	Pedestrians > Cyclists***
Warmth	$F(4, 1124) = 37.18^{***}$	Cyclists > Pedestrians***	Motorists < Pedestrians**	Pedestrians > Motorists***
		Cyclists > Motorists***	Motorists = Cyclists	Pedestrians > Cyclists***
Admiration	$F(4, 1124) = 16.98^{***}$	Cyclists > Pedestrians***	Motorists = Pedestrians	Pedestrians = Motorists
		Cyclists > Motorists***	Motorists = Cyclists	Pedestrians = Cyclists
Contempt	$F(4, 1124) = 41.40^{***}$	Cyclists = Pedestrians	Motorists > Pedestrians*	Pedestrians < Motorists***
		Cyclists < Motorists***	Motorists > Cyclists***	Pedestrians < Cyclists***
Active Facilitation (toward)	$F(4, 1124) = 60.0^{***}$	Cyclists > Pedestrians***	Motorists < Pedestrians***	Pedestrians > Motorists***
		Cyclists > Motorists***	Motorists = Cyclists	Pedestrians > Cyclists***
Passive Facilitation (toward)	$F(4, 1124) = 21.5^{***}$	Cyclists > Pedestrians***	Motorists < Pedestrians**	Pedestrians = Cyclists = Motorists
		Cyclists > Motorists***	Motorists = Cyclists	
Active Harm (toward)	$F(4, 1124) = 14.5^{***}$	Cyclists < Pedestrians[a]	Motorists > Pedestrians***	Pedestrians < Motorists***
		Cyclists < Motorists*	Motorists > Cyclists*	Pedestrians < Cyclists ***
Passive Harm (toward)	$F(4, 1124) = 13.6^{***}$	Cyclists < Pedestrians**	Motorists > Pedestrians*	Pedestrians > Motorists***
		Cyclists = Motorists	Motorists = Cyclists	Pedestrians = Cyclists

Note. ***$p < .001$, **$p < .01$, *$p < .05$; [1]Presentation of relevant post-hoc comparisons for the interaction term. Post-hoc comparisons were calculated using the Holm-Bonferroni correction.

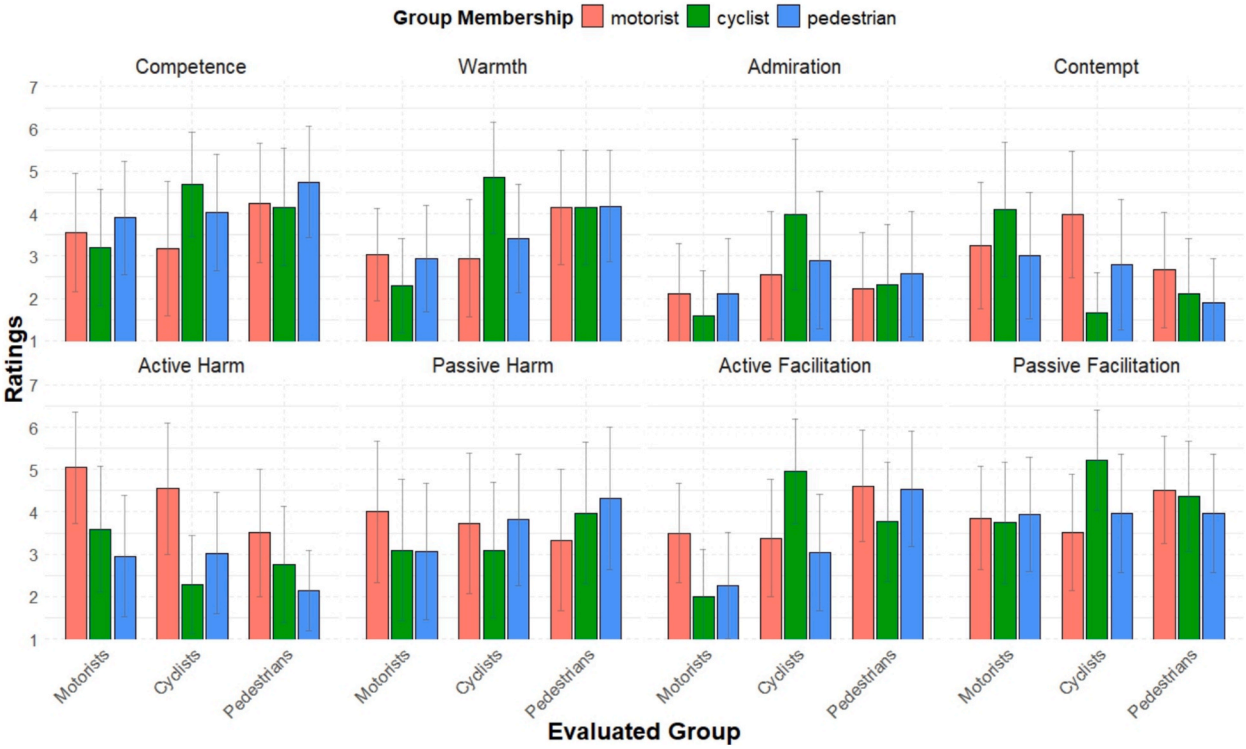

Fig. 2. Mean group ratings on stereotype content and behavioral intentions subscales by group membership.

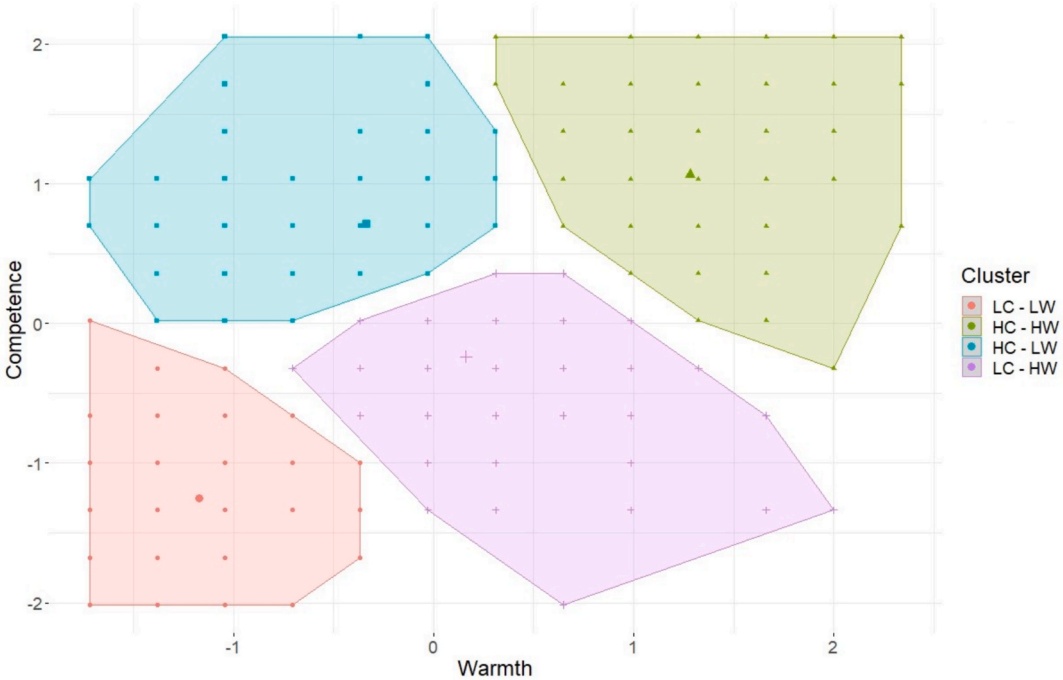

Fig. 3. Cluster representations at the participant level. *Note.* LC = Low Competence; HC = High Competence; LW = Low Warmth; HW = High Warmth.

contrast, *motorist* participants did not exhibit in-group favoritism. Instead, they reported greater active harm toward their own group compared to both outgroups (cyclists and pedestrians). They also expressed higher levels of active and passive facilitation toward pedestrians.

Overall, the results support Hypothesis 3 for pedestrians and cyclists, who perceived their own groups as more competent and warmer and tended to feel more positive emotions toward their ingroup. They also reported stronger facilitation intentions to in-group members and greater intention to harm out-groups. However, this was not the case for motorists, who viewed other drivers more negatively in traffic and exhibited less intentions of facilitation and more intentions of harm towards members of their own group.

3.3. Mapping group perceptions of road user groups

To create a preliminary (and mainly exploratory) map of how different road users perceive themselves, we conducted a cluster analysis in the two-dimensional competence-warmth space. However, unlike previous studies using this approach, our participants only rated three groups ($n = 3$), limiting the ability to generate a comprehensive cartography of road-user perceptions. To address this, we performed the cluster analysis at the participant level rather than the group level, with each data point representing an individual's perception of a specific road-user group (e.g., Borgen & Barnett, 1987). Cluster analyses were performed using *k*-means on standardized warmth and competence ratings. The number of clusters was fixed at four, following the theoretical quadrants of the stereotype content model (Cuddy et al., 2007; Fiske et al., 2002). Stability of the solution was ensured by using 25 random starts (Kaufman & Rousseeuw, 1990). The four-cluster solution closely aligning with those found in SCM-BIAS studies, as depicted in Fig. 3: (1) low competence, low warmth (i.e., negative stereotype), (2) high competence, low warmth (i.e., envious stereotype), (3) low competence, high warmth (i.e., paternalistic stereotype), and (4) high competence, high warmth (i.e., positive stereotype).

K-means analysis further revealed the predominant road-user groups within each cluster. To statistically examine points proportion differences across clusters, we first conducted Fisher's Exact Test for each combination of *User Group* × *Judged Group*. If a significant effect was detected, we then performed pairwise *Z*-tests for proportion differences to identify which clusters were over- or under-represented.[6] This allowed us to establish a ranking and determine where data points were most concentrated. Table 5 presents the detailed distribution of data points across clusters for each user group and judged group, along with proportion test results.

As shown in Table 5, the distribution of points across clusters varied significantly depending on both the user group and the judged group (with the exception of one combination, $p = .13$). Motorist participants primarily categorized themselves within the negative stereotype cluster (41 %), followed by the paternalistic one (30 %). A similar pattern emerged when they evaluated cyclists, with the negative cluster being the most common classification (43 %), followed by paternalistic (30 %). However, when judging pedestrians, motorist participants predominantly placed them in the paternalistic (39 %) and positive (34 %) clusters. Cyclist participants exhibited the most polarized views, primarily classifying motorists in the negative cluster (52 %), while placing the judgements for their own group significantly more in the positive stereotype cluster (56 %). Additionally, cyclist participants were more likely to associate pedestrians with the paternalistic stereotype (46 %). For pedestrian participants, no clear trend emerged regarding their perception of motorists, as they classified them relatively evenly across the negative (29 %), envious (33 %), and paternalistic (26 %) clusters. A similar pattern was observed when evaluating cyclists, with no significant distributional differences across clusters. However, pedestrians judged their own group more favorably, with most responses falling within the positive (42 %) and paternalistic (30 %) stereotype clusters.

4. Discussion

The present study aimed to explore the perceptions and behavioral tendencies among motorists, cyclists, and pedestrians through the lens of the Stereotype Content Model (SCM) and the Behaviors from Intergroup Affect and Stereotype (BIAS) map. Our findings supported the applicability of these models to the traffic environment, shedding light on the stereotypes, emotions, and behavioral intentions that shape interactions among road users, depending on the user's main mode of transportation.

4.1. Social group dynamics on the road

One of the key contributions of this study lies in demonstrating the applicability of the Stereotype Content Model (SCM; Fiske et al., 2002) and the Behaviors from Intergroup Affect and Stereotypes (BIAS) Map (Cuddy et al., 2007) to a novel and underexplored context: road users' interactions. By confirming that stereotype dimensions—warmth and competence—predict behavioral tendencies—namely active and passive facilitation or harm—this research extends the theoretical scope of these widely validated models beyond traditional intergroup domains (e.g., ethnicity, nationality, or occupation), into a situational context of categorization based on modes of transport. This validation is particularly noteworthy because it suggests that road interactions, often perceived as impersonal or anonymous, are in fact socially structured by implicit intergroup perceptions that shape emotions and behavioral intentions.

[6] Fisher's Exact Test was used as an omnibus test of independence. When significant, pairwise differences between clusters were assessed with two-tailed *Z*-tests for independent proportions, with Holm–Bonferroni correction applied to control for multiple comparisons.

Table 5
Points distributions between clusters based on group membership and judged road user group.

User Group	Judged Group	Clusters Positive (n = 269)	Negative (n = 283)	Envious (n = 215)	Paternalistic (n = 358)	Fisher's Exact Test
Motorist	Motorists	9 (7 %)[3]	**51 (41 %)**[1]	27 (22 %)[2]	**37 (30 %)**[1]	$p < .001$
	Cyclists	17 (14 %)[3]	**54 (43 %)**[1]	15 (12 %)[3]	38 (30 %)[2]	$p < .001$
	Pedestrians	**42 (34 %)**[1]	17 (14 %)[2]	17 (14 %)[2]	**48 (39 %)**[1]	$p < .001$
Cyclist	Motorists	4 (3 %)[3]	**59 (52 %)**[1]	26 (23 %)[2]	25 (22 %)[2]	$p < .001$
	Cyclists	**64 (56 %)**[1]	6 (5 %)[3]	9 (8 %)[3]	35 (31 %)[2]	$p < .001$
	Pedestrians	31 (27 %)[2]	16 (14 %)[3]	14 (12 %)[3]	**53 (46 %)**[1]	$p < .001$
Pedestrians	Motorists	16 (11 %)[2]	**40 (29 %)**[1]	**45 (33 %)**[1]	**36 (26 %)**[1]	$p < .001$
	Cyclists	29 (21 %)	29 (21 %)	35 (25 %)	44 (32 %)	$p = .13$
	Pedestrians	**57 (42 %)**[1]	11 (8 %)[3]	27 (20 %)[2]	**42 (30 %)**[1]	$p < .001$

Note. Superscript numbers indicate the cluster's ranking based on pairwise *Z*-tests for proportion differences. Results in bold indicate the most represented clusters.

4.2. In-group favoritism and intergroup perceptions among road users

The results of the present study reveal distinct social dynamics between vulnerable road users (cyclists and pedestrians) and less vulnerable users (motorists). Vulnerable groups demonstrated clear in-group favoritism, expressed through positive stereotypes (i.e., high perceived warmth and competence), emotions (greater admiration), and intentions to facilitate their in-group members, both actively and passively. However, this favoritism did not necessarily extend across vulnerable groups: cyclist participants rated fellow cyclists more positively than pedestrians, and pedestrian participants rated fellow pedestrians more positively than cyclists. This suggests that in-group favoritism was strong within each vulnerable group, but not necessarily between them.

For cyclists, our results support previous findings suggesting that cyclists constitute a cohesive social group, often driven by shared values regarding transportation environmental and health benefits, as well as a collective sense of vulnerability stemming from interactions with motorists (Aldred, 2013; Prati et al., 2017). This shared identity may strengthen perceptions of group belongingness and amplify judgments, emotions, and behaviors that favor in-group members while holding negative perceptions and behavioral intentions toward out-group members.

Interestingly, pedestrians, who are less frequently considered as a social group in the literature (viewed instead as consisting of different groups, e.g., Hu et al., 2021), exhibited a similar pattern of in-group favoritism. In traffic environments, pedestrians face potential conflicts with other modes of transportation, such as cyclists and motorists (Paschalidis et al., 2016), which may foster a sense of shared identity and group belongingness. This may explain the negative judgments and emotional responses toward out-groups while reinforcing positive behavioral intentions toward their in-group in this study.

In contrast to the groups of cyclists and pedestrians, the less vulnerable group of motorists did not display any in-group favoritism. Contrary to our expectations, motorists perceived their own group more negatively than others across stereotype content, emotions, and behavioral intentions. Previous research suggests that motorists tend to show more positive behavioral intentions toward their in-group (Hoekstra et al., 2018), making our findings somewhat unexpected. One possible explanation is that motorists represent a broad and more diverse social group, encompassing various subgroups, leading to a weaker sense of group identity (Balliet et al., 2014). Additionally, societal representations often portray drivers as impatient, reckless, or aggressive, a narrative reinforced by media reports of road rage incidents and conflicts between motorists themselves and with other groups on the road (Cavacuiti et al., 2013; Lupton, 2002). These representations, combined with low group identification, may foster negative expectations and mistrust among motorists, resulting in less favorable in-group perceptions.

These results not only advance theoretical understanding of intergroup perceptions among road users but also have applied implications. They point to the potential of interventions such as public awareness campaigns and perspective-taking training to foster empathy and cooperation across groups. We elaborate further on these practical applications in the Conclusion.

4.3. Agency, morality, and ambivalent stereotypes

Additional insights may be considered from the complementary analysis of perceived agency and morality provided in the supplementary materials (see also Appendix B). Judgments of agency (i.e., the perceived ability to act and influence one's environment; Abele et al., 2016), were highest among motorists (M = 4.66), followed by cyclists (M = 3.43) and pedestrians (M = 3.12). This gradient in favor of motorists may reflects differences in physical control due to vehicle size (Høye, 2019), speed, and infrastructural dominance of the road network. In contrast, morality judgments were higher among cyclists (M = 4.60) and pedestrians (M = 4.49) than motorists (M = 3.18), consistent with compensatory processes described in social identity theory (Tajfel & Turner, 2001), whereby groups in lower-status positions emphasize moral value to maintain a positive social image.

The exploratory mapping of road user perceptions revealed additional complexities. Both motorists and cyclists expressed paternalistic stereotypes toward pedestrians, perceiving them as warm but less competent, worthy of protection yet also an obstacle. Paternalistic stereotypes, characterized by high warmth and low competence, are typically linked to ambivalent behavioral tendencies, such as active facilitation (e.g., helping) combined with passive harm (e.g., hindering).

Altogether, these patterns may suggest a layered structure of social perception: motorists occupying an upper tier, associated with power, dominance, lower morality and also internal ambivalence; cyclists in an intermediate position, asserting some superiority over pedestrians; and pedestrians in the lower tier, viewed as vulnerable, morally positive and paternalized by the other groups. Cluster analysis results further supported this asymmetry, showing that the three groups occupied distinct areas of the competence–warmth space. Overall, the pattern reflects a social ordering of road users, which may shape everyday interactions and judgments in traffic.

4.4. Limitations and further research

This study has several limitations that should be acknowledged. First, the reliance on self-reported data introduces potential biases, including social desirability effects and participants' limited introspective access to their own perceptions and behavioral tendencies. To enhance ecological validity, future research should incorporate behavioral observations, field studies, or experimental paradigms that simulate real-world traffic interactions.

Second, the study was conducted within a single cultural context (France), which may limit the generalizability of the findings. Traffic norms, infrastructure, and cultural attitudes toward road users can vary substantially across countries (Nordfjærn et al., 2014; Öztürk et al., 2025; Uzondu et al., 2020). Cross-cultural comparisons would be valuable in testing the robustness of the observed intergroup dynamics and the applicability of the SCM and BIAS Map models in other sociocultural environments. Furthermore, another limitation concerns the sociodemographic profile of our sample, which included a relatively high proportion of well-educated participants. Education level has been associated with stronger pro-environmental attitudes (Meyer, 2015), which could influence perceptions of road users. For instance, well-educated individuals may be more inclined to express more critical views toward car use (independent of their means of transportation), in line with broader societal concerns about pollution generated by individual cars. Future research should recruit more socio-demographically diverse samples to examine whether our findings generalize across different education levels.

Third, while this study focused on three major categories of road users—motorists, cyclists, and pedestrians—it did not account for the increasing diversity in urban mobility. The growing presence of e-scooter riders, delivery personnel, and other emerging micromobility users may introduce novel group boundaries and stereotype dynamics. Future research should explore how these new categories are perceived and how they interact with existing road-user groups.

Finally, while our data suggested a potential hierarchical structure among road users, this interpretation remains exploratory and requires further empirical investigation to establish its robustness.

5. Conclusion and practical implications

The present study underscores the importance of social perception in shaping road user interactions, with the SCM and BIAS map offering valuable frameworks for understanding these dynamics. By shedding light on the stereotypes, emotions, and behaviors at play, this research paves the way for interventions that promote safer and more cooperative road environments for all.

Understanding the stereotypes and emotions that shape road user interactions can inform the development of targeted interventions to improve road safety and cooperation. For instance, public awareness campaigns could challenge negative stereotypes by emphasizing shared vulnerabilities (e.g., all road users can be injured in accidents) and highlighting common goals. Educational modules in driving schools or even at earlier stages (e.g., elementary or secondary school) could raise awareness about the existence of stereotypes toward different road-user groups, an important first step in reducing intergroup bias. Interventions based on perspective-taking have proven effective in reducing intergroup prejudice (Galinsky & Moskowitz, 2000; Todd & Galinsky, 2014). Such trainings could draw on established approaches like relational frame theory or be enhanced by technological tools such as virtual reality, which allows participants to embody the perspective of another group (Tassinari et al., 2022). By enabling motorists, cyclists, or pedestrians to "step into each other's shoes," these immersive experiences may foster empathy and reduce antagonism.

Finally, urban planning strategies that prioritize inclusive infrastructure, such as clearly marked cycling lanes, safe pedestrian crossings, and traffic-calmed areas, could reduce perceived competition over road space and may mitigate intergroup tensions. Designing spaces that encourage encounters under positive conditions may also increase mutual understanding, thereby reinforcing empathy across groups.

CRediT authorship contribution statement

Clement F.A. Andrieu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aurélie Dommes:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Julie Collange:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Item selection and translation methodology.

First, a bilingual member of the research team proposed a preliminary version in French adapted to the road environment. Then, the other two authors, two experts in the field of traffic psychology and social psychology, each native French speakers but fluent in English, proposed corrections. The third step was to arrive at an agreed French translation based on the three viewpoints, with the aim of staying as close as possible to the definition of the authors of the original English scales and adapting it to the road environment while remaining meaningful in French. Relevance (r) to the road environment and rating of each translation (t) were assessed by rating each item on a scale from 1 (not relevant at all) to 5 (strongly relevant). For factors not present in the original scale (agency and morality), three items for each factor were chosen from Abele et al.'s (2016) study (self-confident, purposeful, and feeling superior for agency; honest, reliable, and trustworthy for morality). With three authors, a threshold was set at Mr $\geq$ 4.00 and SDr $\leq$ 1.00 to indicate a high inter-judge agreement. If an item or its translation was deemed not appropriate, a new translation was proposed by a second author and rated anew by all three, and so on until a strong inter-judge agreement was obtained (M $\geq$ 4.00 and SD $\leq$ 1.00).

Details regarding the final item selection and difference between original measurements.

Items for the competence and warmth subscales (stereotype content) were consistent with those originally developed by Cuddy et al. (2007) in the BIAS map. These items were rated highly relevant to their applicability in the traffic context and there were high levels of agreement regarding their translation into French. To maintain consistency in the number of items across all subscales, the two items with the highest agreement ratings for agency (*self-confident* and *feel superior*) and morality (*reliable* and *trustworthy*) in relation to traffic were retained for the study. Consequently, the items *purposeful* ($M_r = 3.33$) and *honest* ($M_r = 4.00$) were excluded.

Items comprising the contempt, pity, and envy subscales were the same as in the original BIAS map (Cuddy et al., 2007), demonstrating a high level of inter-judge agreement regarding their relevance to traffic situations, as well as for their French translation. In contrast, regarding the admiration subscale, while the item *admire* was deemed highly relevant in the traffic context, *proud* was not ($M_r = 3.33$). Therefore, we proposed to replace it with the item *respect* which is used in social perception studies to measure admiration (Algoe & Haidt, 2009; Onu et al., 2016). The item *respect* received high ratings for relevance in the traffic context and was consequently retained in place of *proud*.

Items for both active and passive facilitation were identical to those from Cuddy et al.'s (2007) original BIAS map scale and demonstrated high inter-judge agreement regarding their relevance to traffic. However, the straightforward French translations for the items *protect* (active facilitation) and *associate with* (passive facilitation) were deemed unsatisfactory ($M_t = 2.67$ for both items). Consequently, new translations were proposed while ensuring the preservation of the original items' meaning. *Protect* was translated as '*se montrer prévenant*' ("being considerate," $M_t = 4.67$), and *associate with* as '*faire preuve de bonne volonté*' ("show good will," $M_t = 5.00$). The two items comprising the active harm subscale are identical to those found in the original BIAS map scale and were rated highly relevant to the road environment. However, literal French translations were deemed inadequate due to the risk of social desirability bias in reporting counter-normative behaviors such as road *attacks* and *fights* ($M_t = 3.00$ and $M_t = 3.67$, respectively). New translations for active harm items were then proposed to mitigate social desirability while preserving the meaning of the original item. *Fight* was translated to '*insulter*' ("insult, verbally spar," $M_t = 4.66$), and *attack* as '*se montrer aggressif*' ("to display aggressivity," $M_t = 5.00$). Regarding passive harm, the original BIAS map item *exclude* was deemed highly relevant for the road environment, while *demean* was not ($M_r = 3.67$). This item was replaced by *hinder*, a previously pre-tested item in the study by Cuddy et al. (2007), though it was not retained by them due to their questionnaire being too long. *Hinder* and its translation were deemed highly appropriate.

Final SCM-BIAS map item pool and inter-judge agreement score regarding relevance, their French translations, and the agreements regarding their translations.

Sub-scale	Item	Mean (*SD*) inter-judge relevance rating	French translation	Mean (*SD*) inter-judge rating for the final translation
Competence	Competent	5 (0.00)	Compétents	5 (0.00)
	Capable	4.33 (0.57)	Capables	4.33 (0.57)
Warmth	Warm	5 (0.00)	Chaleureux	5 (0.00)
	Friendly	5 (0.00)	Sympathiques	5 (0.00)
Agency	Self-confident	5 (0.00)	Sûrs d'eux	4.66 (0.57)
	Feel superior	5 (0.00)	Se sentant supérieurs	5 (0.00)
Morality	Reliable	5 (0.00)	Fiables	5 (0.00)
	Trustworthy	5 (0.00)	Dignes de confiance	5 (0.00)
Contempt	Contempt	5 (0.00)	Du mépris	5 (0.00)
	Disgust	5 (0.00)	Du dégoût	5 (0.00)
Pity	Pity	5 (0.00)	De la pitié	5 (0.00)

(*continued on next page*)

(*continued*)

Sub-scale	Item	Mean (*SD*) inter-judge relevance rating	French translation	Mean (*SD*) inter-judge rating for the final translation
	Sympathy	5 (0.00)	De la compassion	5 (0.00)
Admiration	Admire	4.66 (0.57)	De l'admiration	5 (0.00)
	Respect	4.33 (0.57)	Du respect	5 (0.00)
Envy	Envious	4.66 (0.57)	De l'envie	4.66 (0.57)
	Jealous	5 (0.00)	De la jalousie	4.66 (0.57)
Active facilitation	Help	5 (0.00)	Les aider	5 (0.00)
	Protect	5 (0.00)	Se montrer prévenant	4.66 (0.57)
Active harm	Fight	5 (0.00)	Les insulter	4.66 (0.57)
	Attack	5 (0.00)	Se montrer agressif	5 (0.00)
Passive facilitation	Cooperate with	4.66 (0.57)	Coopérer	4.33 (0.57)
	Associate with	4.33 (0.57)	Faire preuve de bonne volonté	5 (0.00)
Passive harm	Exclude	5 (0.00)	Les ignorer	5 (0.00)
	Hinder	5 (0.00)	Les gêner	5 (0.00)

Appendix B

Descriptive statistics by category of road user.

Table B1
Descriptive statistics for driver participants.

	Motorists				Cyclists				Pedestrians			
	M	*SD*	Min	Max	*M*	*SD*	Min	Max	*M*	*SD*	Min	Max
Competence	3.56	1.39	1.00	7.00	3.17	1.59	1.00	7.00	4.25	1.40	1.00	7.00
Warmth	3.03	1.09	1.00	6.00	2.95	1.39	1.00	7.00	4.15	1.35	1.00	7.00
Agency	4.66	1.59	1.00	7.00	4.74	1.86	1.00	7.00	3.47	1.74	1.00	7.00
Morality	3.18	1.29	1.00	7.00	2.74	1.32	1.00	6.50	3.67	1.38	1.00	7.00
Admiration	2.10	1.18	1.00	6.00	2.55	1.51	1.00	7.00	2.23	1.33	1.00	7.00
Contempt	3.25	1.49	1.00	7.00	3.97	1.50	1.00	7.00	2.67	1.36	1.00	6.00
Pity	2.01	1.04	1.00	5.00	2.30	1.37	1.00	6.00	2.52	1.53	1.00	7.00
Envy	2.49	1.28	1.00	5.50	1.95	1.01	1.00	5.00	1.74	0.92	1.00	5.50
AF	3.50	1.17	1.00	6.00	3.38	1.38	1.00	7.00	4.61	1.31	1.00	7.00
PF	3.85	1.22	1.50	6.50	3.51	1.36	1.00	7.00	4.52	1.27	1.00	7.00
AH	5.05	1.32	1.00	7.00	4.54	1.56	1.00	7.00	3.51	1.51	1.00	7.00
PH	4.00	1.66	1.00	7.00	3.73	1.65	1.00	7.00	3.33	1.68	1.00	7.00

Note. AF = Active Facilitation. PF = Passive Facilitation. AH = Active Harm. PH = Passive Harm.

Table B2
Descriptive statistics for cyclist participants.

	Motorists				Cyclists				Pedestrians			
	M	*SD*	Min	Max	*M*	*SD*	Min	Max	*M*	*SD*	Min	Max
Competence	3.20	1.37	1.00	6.00	4.69	1.23	1.00	7.00	4.15	1.38	1.00	7.00
Warmth	2.30	1.11	1.00	6.00	4.85	1.31	1.00	7.00	4.14	1.36	1.00	7.00
Agency	5.89	1.35	1.00	7.00	3.43	1.64	1.00	7.00	2.99	1.64	1.00	7.00
Morality	2.51	1.14	1.00	6.50	4.60	1.33	1.00	7.00	3.62	1.39	1.00	7.00
Admiration	2.26	0.99	1.00	5.00	3.97	1.77	1.00	7.00	2.32	1.42	1.00	7.00
Contempt	4.10	1.58	1.00	7.00	1.66	0.95	1.00	5.50	2.11	1.30	1.00	6.00
Pity	2.34	1.55	1.00	7.00	2.09	1.33	1.00	6.00	2.52	1.53	1.00	6.00
Envy	2.10	1.39	1.00	6.50	2.32	1.19	1.00	5.50	1.63	0.89	1.00	5.00
AF	1.99	1.11	1.00	5.50	4.96	1.22	1.00	7.00	3.76	1.42	1.00	7.00
PF	3.75	1.44	1.00	6.50	5.21	1.18	1.00	7.00	4.36	1.30	1.00	7.00
AH	3.60	1.48	1.00	7.00	2.28	1.17	1.00	7.00	2.75	1.37	1.00	7.00
PH	3.10	1.67	1.00	7.00	3.10	1.60	1.00	7.00	3.97	1.66	1.00	7.00

Note. AF = Active Facilitation. PF = Passive Facilitation. AH = Active Harm. PH = Passive Harm.

Table B3
Descriptive statistics for pedestrian participants.

	Motorists				Cyclists				Pedestrians			
	M	*SD*	Min	Max	*M*	*SD*	Min	Max	*M*	*SD*	Min	Max
Competence	3.90	1.34	1.00	7.00	4.03	1.37	1.00	7.00	4.74	1.31	1.00	7.00
Warmth	2.94	1.25	1.00	6.00	3.41	1.28	1.00	6.50	4.18	1.32	1.00	7.00
Agency	5.31	1.50	1.00	7.00	4.57	1.79	1.00	7.00	3.12	1.59	1.00	7.00
Morality	3.27	1.28	1.00	6.00	3.45	1.29	1.00	7.00	4.49	1.31	1.00	7.00
Admiration	2.10	1.31	1.00	7.00	2.89	1.62	1.00	7.00	2.57	1.48	1.00	7.00
Contempt	3.00	1.49	1.00	6.50	2.80	1.53	1.00	7.00	1.90	1.04	1.00	5.00
Pity	1.94	1.20	1.00	6.00	2.04	1.23	1.00	7.00	2.01	1.21	1.00	6.00
Envy	2.62	1.42	1.00	6.00	2.41	1.38	1.00	6.00	1.74	1.02	1.00	4.50
AF	2.25	1.26	1.00	7.00	3.05	1.37	1.00	6.50	4.54	1.37	1.00	7.00
PF	3.94	1.35	1.00	7.00	3.97	1.40	1.00	7.00	3.97	1.39	1.00	7.00
AH	2.96	1.44	1.00	7.00	3.03	1.44	1.00	7.00	2.14	0.94	1.00	5.00
PH	3.07	1.61	1.00	6.00	3.82	1.55	1.00	7.00	4.31	1.68	1.00	7.00

Note. AF = Active Facilitation. PF = Passive Facilitation. AH = Active Harm. PH = Passive Harm.

Appendix C. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.trf.2025.103419.

Data availability

OSF depository link is present in the manuscript with data files.

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